THE JACOBIAN DETERMINANT ANNIHILATES THE NAIVE COTANGENT COMPLEX OF A PRESENTATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof transfers the vanishing of multiplication by the Jacobian determinant from the matrix complex K through the two-term presentation complex L up to Ext^1 of the full naive cotangent complex via a distinguished triangle

Editorial note. The label lemma-presentation-rig-smooth names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the naive cotangent complex NL of a ring map and Ext groups over D(B)
- the two-term complex representing NL of a quotient presentation (Algebra, Section algebra-section-netherlander)
- the More on Algebra lemmas: the determinant of the Jacobian matrix acts trivially on the displayed matrix complex (more-algebra-lemma-silly) and a surjection of two-term complexes transfers vanishing of multiplication (more-algebra-lemma-two-term-surjection-map-zero)
- distinguished triangles in the derived category and the long exact Ext sequence

Source reference. Original section 00S0: sources/stacks-chow/tags.txt, source lines 1022–1022. Prerequisite pointer invoked at proof line(s) 1617; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0G9L: sources/stacks-chow/tags.txt, source lines 20306–20306. Prerequisite pointer invoked at proof line(s) 1613; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0G9I: sources/stacks-chow/tags.txt, source lines 20303–20303. Prerequisite pointer invoked at proof line(s) 1627; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let A be a ring. Let $f_1, \ldots, f_m \in A[x_1, \ldots, x_n]$ and set $B = A[x_1, \ldots, x_n]/(f_1, \ldots, f_m)$. Assume $m \leq n$ and set $g = \det_{1 \leq i,j \leq m}(\partial f_j / \partial x_i)$. Then*

1. *g annihilates $\mathrm{Ext}^1_B(NL_{B/A}, N)$ for every B-module N,*
2. *if $n = m$, then multiplication by g on $NL_{B/A}$ is 0 in $D(B)$.*

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Proof. Let T be the $m \times m$ matrix with entries $\partial f_j/\partial x_i$ for $1 \leq i, j \leq n$. Let $K \in D(B)$ be represented by the complex $T : B^{\oplus m} \to B^{\oplus m}$ with terms sitting in degrees -1 and 0. By More on Algebra, Lemmas 0G9L we have $g : K \to K$ is zero in $D(B)$. Set $J = (f_1, \ldots, f_m)$. Recall that $NL_{B/A}$ is homotopy equivalent to $J/J^2 \to \bigoplus_{i=1,\ldots,n} B\mathrm{d}x_i$, see Algebra, Section 00S0. Denote L the complex $J/J^2 \to \bigoplus_{i=1,\ldots,m} B\mathrm{d}x_i$ to that we have the quotient map $NL_{B/A} \to L$. We also have a surjective map of complexes $K \to L$ by sending the jth basis element in the term $B^{\oplus m}$ in degree -1 to the class of f_j in J/J^2. Picture

$$NL_{B/A} \to L \leftarrow K$$

From More on Algebra, Lemma 0G9I we conclude that multiplication by g on L is 0 in $D(B)$. On the other hand, the distinguished triangle $B^{\oplus n-m}[0] \to NL_{B/A} \to L$ shows that $\mathrm{Ext}^1_B(L, N) \to \mathrm{Ext}^1_B(NL_{B/A}, N)$ is surjective for every B-module N and hence annihilated by g. This proves part (1). If $n = m$ then $NL_{B/A} = L$ and we see that (2) holds. $\square$